

SruthiScribe

Sing it. Read it back as svaras.

A browser-based Carnatic transcription engine, an open machine-readable library of the repertoire, and guided practice against it.

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THE PROBLEM

Carnatic music is taught by ear, and stays there

A student cannot see what they just sang

Correction happens in the lesson, from the teacher's ear. Between lessons there is no feedback at all.

Notation exists, but not where you need it

Printed books, PDFs and scanned archives — none of it queryable, none of it aligned to what you are singing.

The reference corpus is scattered and closed

Ragam definitions on one site, lyrics on another, notation in a 1904 book. Little of it is machine-readable.

What a learner actually has today

A tanpura app for the drone.

A tuner that says whether one note is sharp.

A PDF of notation they cannot search.

A recording of their own practice they will never listen back to.

Nothing that turns the singing into notation.

THE NEED

A loop, not a tool

1 • Hear yourself

Sing a phrase; get back svaras with sthayi, gamaka and avartana bars, laid out the way the tradition prints it.

2 • Find it written

Compositions, ragams, talams and notation in one place, searchable, with the source and licence of every row recorded.

3 • Practise against it

Sing a written composition and be told, svara by svara, which you held and which you missed — in cents, not opinion.

Each stage feeds the next. The library gives the decoder its ragam and tala; the decoder gives the library new readings; the notation gives practice something exact to score against. A student can sing a kriti they cannot yet write, contribute it, and then practise the thing they just wrote.

What it does today

Record or upload, and read it back

Microphone or file, in the browser. Svaras with octave, duration, gamaka class and cents deviation.

Tala-aware layout

All seven suladi sapta talas × five jatis, nadai 1x–8x, avartana and anga bar lines, sahitya under the svaras.

Fix what it got wrong

Edit any svara, mark the eduppu, regroup speeds — the decode is a starting point, not a verdict.

Export

Print-quality PDF laid out as the tradition prints it, JSON, or contribute the reading to the library.

A library of 598 compositions

42 with complete notation, searchable by ragam, tala, composer, deity and how complete the notation is.

939 ragams

All 72 melakartas and their janyas, with arohana/avarohana; 120 wired into the decoder as constraints.

Provenance on every row

Source, licence and page, so a notation can be checked against the book it came from.

Sadhana — guided practice

Sing any notated composition against a scrolling stage with a drone and a tala click; scored per svara in cents.

A signal-processing pipeline, entirely in the browser



No server, no model, no key

The engine is 46 kB of dependency-free JavaScript. Transcription works offline; nothing is uploaded.

The ragam is the decoder's state space

Viterbi over that ragam's svarasthanas, penalising switches — so the decode cannot invent a note outside the ragam.

Postgres + REST for the library

Supabase; the page talks to it directly. Notation stored as JSON with the tala and akshara grid.

The harness scores the shipping code

tools/eval.js extracts the engine from the page itself and scores it against public datasets — the two cannot drift.

MEASURED, NOT CLAIMED

Accuracy on studio recordings

91.7%

svara accuracy
(label + octave)

82.5%

raw pitch accuracy

2.5%

octave error rate

3.2¢

tonic (sruthi) error

Scored on 33 items of Sanidha (Georgia Tech, CC BY 4.0), studio multitrack, against the dataset's own pitch annotations. It declines 27% of sung frames rather than guess them.

How the measurement is kept honest

The harness runs the engine that ships, not a copy. Every change must improve two datasets (Sanidha and Saraga), both halves of a holdout split, and survive a deliberately mis-set tonic before it lands — what helps one corpus and hurts the other is rejected. Annotation faults are handled in the open: three records whose contour tracks another instrument are excluded, and frames the dataset itself doubled an octave are repaired by rules that read only the reference, never the engine's output. The web page publishes these same figures, and a test fails if they drift from the measurement.

Rescuing a 1904 book by machine

Subbarama Dikshitar's Sangita Sampradaya Pradarsini (1904) is the only public-domain source of complete Carnatic notation — and it has never been machine-readable.

598

compositions

42

with complete notation

939

ragams

4,695

svaras extracted from SSP

The extractor reads the book's own notation scheme from its page iii — capitals are two aksharas, underlines halve, dots mark sthayi — and verifies every section against the tala the book prints for it. 26 kirtanas parse clean end to end; 12 of them were confirmed independently against a separate catalogue, agreeing on both title and ragam. Nothing that fails verification is loaded.

LIMITATIONS

What it cannot do yet

Ragam identification is weak — 15% top-1

A pitch histogram cannot separate ragams that share a svara set and differ by phrase. The user still has to name the ragam.

Gamaka classification is coarse

Five machine labels, not the dasavidha taxonomy. It says 'kampita', not which kampita.

It abstains, and it false-alarms

27% of sung frames get no svara — declined, not guessed — and 32% of silences still get called a note.

One voice, no ensemble

Trained on and scored against solo vocal. Accompaniment in the room degrades it.

The library is mostly metadata

433 of 598 rows have no notation. Complete notation is not openly licensed at scale; that is a licensing wall, not an engineering one.

The SSP extractor verifies 44% of sections

Errors are per-glyph now — underline geometry, tight spacing. Each further point is slow work.

No Tyagaraja notation

SSP covers Dikshitar. The Walajapet manuscripts are scans with no text layer.

Not a substitute for a teacher

It measures pitch. It does not hear bhava, and it should not be sold as if it did.

What is genuinely new here

Already exists

- Tuners that name the svara you are holding — Carnatic Tuner, Shruti.
- Practice apps with pitch feedback and lesson tracks — Riyaz, Carnatic Singer. Sadhana is our answer to these, and they got there first.
- AI accompaniment and raga detection — NaadSadhana, an Apple Design Award winner with 410 ragas.
- Notation typing tools — Haathi Carnatic Editor.
- Research corpora — CompMusic, Saraga, Sanidha.

Less crowded

- Continuous transcription of a phrase into tala-aligned notation, not just the current note.
- Runs entirely in the browser — no install, no account, works offline.
- Published accuracy against public datasets, with the harness in the repo.
- Practice scored against an open library rather than a proprietary lesson set — including a 1904 book nobody else can drill you on.
- **A machine-readable SSP corpus with provenance — as far as we know, the first.**

Honest reading: transcription is differentiated, not unique, and practice apps had a decade's head start. What none of them close is the loop — sing something down, contribute it, then be drilled on it. The extracted SSP corpus is what makes that loop worth closing.

WHERE IT GOES

Next

- Raise the SSP yield — every point converts directly into complete works.
- Cut the voicing false alarm; it is the weakest measured number.
- Phrase-level ragam identification, where the published work actually lives.
- Attach openly-licensed audio to notated works, so a student can hear and read together.

Take Sadhana from pitch to timing — holding a svara for the right number of aksharas is half of singing in tala.
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